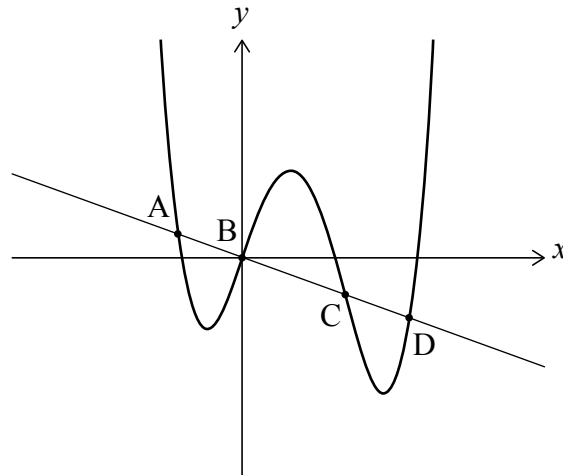


2. [Maximum mark: 28]

This question investigates a ratio of lengths found from the line passing through the points of inflexion of a quartic curve of the form $y = x^4 - mx^3 + nx$.

The curve $y = x^4 - 3x^3 + 3x$ has points of inflexion at B and C. The line passing through B and C intersects the curve again at points A and D. This is shown in the following graph.



- (a) Find $\frac{d^2y}{dx^2}$. [3]
- (b) Find the coordinates of B and C. [4]
- (c) Show that the equation of the line through B and C is $y = -0.375x$. [2]
- (d) Find, correct to three decimal places, the x -coordinate of D. [2]

Now consider the general curve $y = x^4 - mx^3 + nx$, where $m, n \in \mathbb{R}$ and $m > 0$.

- (e) Find the x -coordinates of the two points of inflexion in terms of m . [3]

Let these points of inflexion be B and C. The line passing through B and C intersects the curve again at points A and D. Let x_A be the x -coordinate of point A, and similarly for x_B , x_C and x_D . It is given that $x_A < x_B < x_C < x_D$.

- (f)
 - (i) Write down the coordinates of B. [1]
 - (ii) Find, in terms of m and n , the coordinates of C. [2]

(This question continues on the following page)

(Question 2 continued)

(g) Show that the equation of the line through B and C is $y = \left(-\frac{m^3}{8} + n \right) x$. [2]

(h) Show that $x_A = \frac{m}{4} - \frac{m}{4}\sqrt{5}$. [7]

(i) Hence, find the exact value of $\frac{x_B - x_A}{x_C - x_B}$. [2]
